1. Show records of 'male' patient from 'southwest' region.

```
USE subqueries;

SELECT * FROM insurance_data
WHERE gender='male' and region='southwest';
```

index	PatientID	age	gender	bmi	bloodpressure	diabetic	children	smoker	region	claim
16	17	35	male	34.1	90	No	0	No	southwest	1261.44
17	18	41	male	34.4	84	No	0	No	southwest	1261.86
18	19	49	male	35.4	97	Yes	0	No	southwest	1263.25
21	22	34	male	31.1	96	No	0	No	southwest	1526.31
47	48	31	male	39.5	80	Yes	0	No	southwest	1682.6
72	73	22	male	34.4	98	Yes	0	No	southwest	1826.84
73	74	20	male	20.9	98	No	1	No	southwest	1832.09
74	75	50	male	24.6	90	No	1	No	southwest	1837.24
76	77	38	male	28.4	87	No	1	No	southwest	1842.52
84	85	36	male	22	88	Yes	1	No	southwest	1964.78
100	101	30	male	22.3	83	Yes	1	No	southwest	2103.08
140	141	47	male	35.2	80	No	1	No	southwest	2416.96

2. Show all records having bmi in range 30 to 45 both inclusive.

```
USE subqueries;

SELECT * FROM insurance_data
WHERE bmi between 30 and 45;
```

index	PatientID	age	gender	bmi	bloodpressure	diabetic	children	smoker	region	claim
1	2	24	male	30.1	87	No	0	No	southeast	1131.51
2	3		male	33.3	82	Yes	0	No	southeast	1135.94
3	4		male	33.7	80	No	0	No	northwest	1136.4
4	5		male	34.1	100	No	0	No	northwest	1137.01
5	6		male	34.4	96	Yes	0	No	northwest	1137.47
6	7		male	37.3	86	Yes	0	No	northwest	1141.45
7	8	19	male	41.1	100	No	0	No	northwest	1146.8
8	9	20	male	43	86	No	0	No	northwest	1149.4
15	16	32	male	30.4	86	Yes	0	No		1256.3
16	17	35	male	34.1	90	No	0	No	southwest	1261.44
17	18	41	male	34.4	84	No	0	No	southwest	1261.86
18	19	49	male	35.4	97	Yes	0	No	southwest	1263.25

3. Show minimum and maximum bloodpressure of diabetic patient who smokes. Make column names as MinBP and MaxBP respectively

```
USE subqueries;

SELECT MAX(bloodpressure) AS 'MAXBP' , MIN(bloodpressure) AS 'MINBP'
FROM insurance_data
WHERE smoker='yes';
```

MAXBP	MINBP
140	80

4. Find no of unique patients who are not from southwest region.

```sql
USE subqueries;

SELECT DISTINCT PatientID
FROM insurance_data
WHERE region not in ('southwest');
```

PatientID
1
2
3
4
5
6
7
8
9
10

5. Total claim amount from male smoker.

```sql
USE subqueries;

SELECT SUM(claim) AS 'TOTAL CLAIM AMOUNT'
FROM insurance_data
WHERE gender='male';
```

TOTAL CLAIM AMOUNT
9437617.93

6. Select all records of south region.

```sql
USE subqueries;

SELECT * FROM insurance_data
WHERE region LIKE 'south%';
```

index	PatientID	age	gender	bmi	bloodpressure	diabetic	children	smoker	region	claim
0	1	39	male	23.2	91	Yes	0	No	southeast	1121.87
1	2	24	male	30.1	87	No	0	No	southeast	1131.51
2	3		male	33.3	82	Yes	0	No	southeast	1135.94
16	17	35	male	34.1	90	No	0	No	southwest	1261.44
17	18	41	male	34.4	84	No	0	No	southwest	1261.86
18	19	49	male	35.4	97	Yes	0	No	southwest	1263.25
19	20	48	male	33.3	91	Yes	0	No	southeast	1391.53
20	21	45	male	23.2	85	Yes	0	No	southeast	1515.34
21	22	34	male	31.1	96	No	0	No	southwest	1526.31
22	23	18	male	35.5	100	Yes	0	No	southeast	1532.47

7. No of patient having normal blood pressure. Normal range[90-120]

```sql
USE subqueries;

SELECT COUNT(PatientID)
FROM insurance_data
WHERE bloodpressure >=90 and bloodpressure <=120;
```

COUNT(PatientID)
758

8. No of patient below 17 years of age having normal blood pressure as per below formula - BP normal range = 80+(age in years × 2) to 100 + (age in years × 2)

```sql
USE subqueries;

SELECT *
FROM insurance_data
WHERE age<17 AND bloodpressure BETWEEN (80 + (age * 2)) AND (100 + (age * 2));
```

index	PatientID	age	gender	bmi	bloodpressure	diabetic	children	smoker	region	claim
2	3		male	33.3	82	Yes	0	No	southeast	1135.94
3	4		male	33.7	80	No	0	No	northwest	1136.4
4	5		male	34.1	100	No	0	No	northwest	1137.01
5	6		male	34.4	96	Yes	0	No	northwest	1137.47
6	7		male	37.3	86	Yes	0	No	northwest	1141.45

9. What is the average claim amount for non-smoking female patients who are diabetic?

```sql
USE subqueries;

SELECT ROUND(AVG(claim),2)
FROM insurance_data
WHERE gender='female' AND smoker='No' AND diabetic='Yes';
```

ROUND(AVG(claim),2)
8593.77

10. Write a SQL query to update the claim amount for the patient with PatientID = 1234 to 5000.

SQL File 3*

```sql
USE subqueries;

UPDATE campusx.insurance_data SET
claim=5000 WHERE PatientID = 1234;
SELECT * FROM insurance_data WHERE PatientID = 1234;
```

Result Grid

index	PatientID	age	gender	bmi	bloodpressure	diabetic	children	smoker	region	claim
1233	1234	20	male	33.6	103	No	1	Yes	northeast	37607.53

11. Write a SQL query to delete all records for patients who are smokers and have no children

```sql
USE subqueries;

DELETE FROM insurance_data
WHERE smoker='Yes' AND children=0;
SELECT * FROM insurance_data WHERE children=0;
```

Result Grid

index	PatientID	age	gender	bmi	bloodpressure	diabetic	children	smoker	region	claim
0	1	39	male	23.2	91	Yes	0	No	southeast	1121.87
1	2	24	male	30.1	87	No	0	No	southeast	1131.51
2	3		male	33.3	82	Yes	0	No	southeast	1135.94
3	4		male	33.7	80	No	0	No	northwest	1136.4
4	5		male	34.1	100	No	0	No	northwest	1137.01
5	6		male	34.4	96	Yes	0	No	northwest	1137.47
6	7		male	37.3	86	Yes	0	No	northwest	1141.45
7	8	19	male	41.1	100	No	0	No	northwest	1146.8
8	9	20	male	43	86	No	0	No	northwest	1149.4